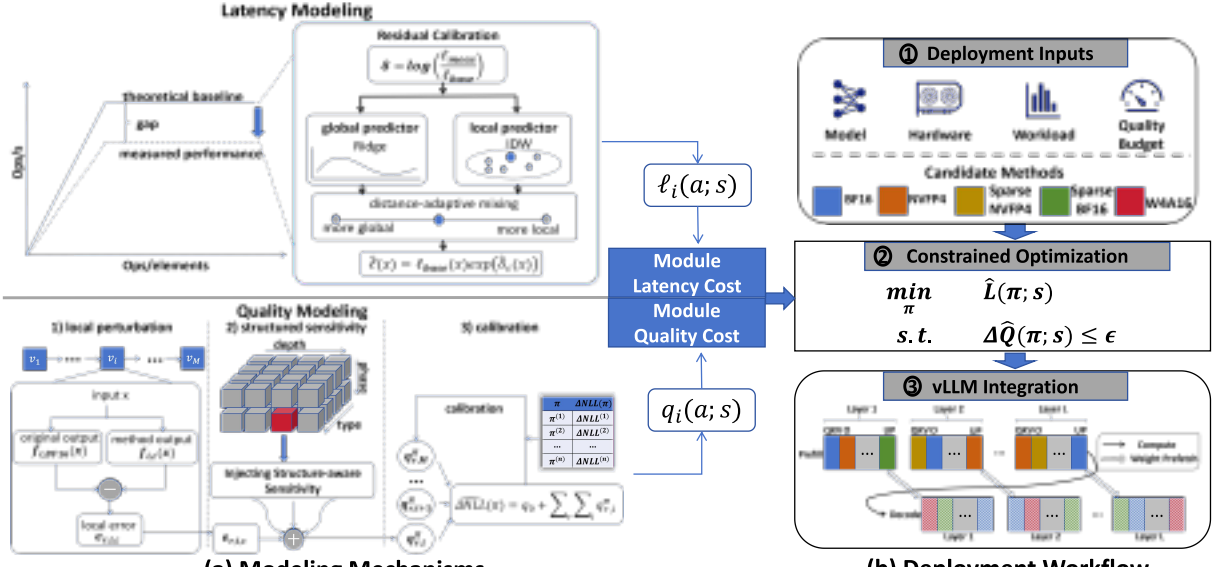

Latency Modeling
Residual Calibration
theoretical baseline
gap
measured performance
Ops/s
Ops/elements
global predictor
Ridge
local predictor
IDW
distance-adaptive mixing
more global
more local
Quality Modeling
1) local perturbation
2) structured sensitivity
depth
input
type
3) calibration
input x
original output
method output
local error
Injecting Structure-aware Sensitivity
calibration
Module Latency Cost
Module Quality Cost
① Deployment Inputs
Model
Hardware
Workload
Quality Budget
Candidate Methods
BF16
NVFP4
Sparse NVFP4
Sparse BF16
W4A16
② Constrained Optimization
③ vLLM Integration
Compute
Weight Prefetch
(a) Modeling Mechanisms
(b) Deployment Workflow